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Quantum annealing learning search for solving QUBO problems

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Abstract
In this paper, we present a novel strategy to solve optimization problems within a hybrid quantum-classical scheme based on quantum annealing, with a particular focus on QUBO problems. The proposed algorithm implements an iterative structure where the representation of an objective function into the annealer architecture is learned and already visited solutions are penalized by a tabu-inspired search. The result is a heuristic search equipped with a learning mechanism to improve the encoding of the problem into the quantum architecture. We prove the convergence of the algorithm to a global optimum in the case of general QUBO problems. Our technique is an alternative to the direct reduction of a given optimization problem into the sparse annealer graph.

Keywords Quantum annealing · Optimization problems · Tabu search

1 Introduction

Quantum annealing (QA) is a type of heuristic search to solve optimization problems [6,13,14]. The solution to a given problem corresponds to the ground state of a quantum system with total energy described by a *problem Hamiltonian* H_P that is a self-adjoint operator on the Hilbert space where the considered quantum system is described. The main idea is to set a time evolution in the interval $[0, T]$ given by a time-dependent

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Hamiltonian $\{H(t)\}_{t\in[0,T]}$ such that $H(T) = H_P$ providing that the quantum system ends in the ground state of H_P with high probability. In QA, the time-dependent Hamiltonian of the considered quantum system (typically a n-qubit system) is:

$$H(t) = H_P + \gamma(t)H_D, \tag{1}$$

where $\gamma : [0, T] \to \mathbb{R}$ is a smooth monotone function decreasing to 0. H_D is the so-called *disorder Hamiltonian* which does not commute with H_P, and it introduces the quantum fluctuations to permit the heuristic search escaping local minima [14]. The annealing procedure is implemented decreasing the kinetic contribution of H_D. The difference between QA and simulated annealing (SA) is that QA exploits the *tunnel effect* to escape local minima instead of *thermal hill-climbing*.

QA can be physically realized considering a quantum spin glass that is a network of qubits arranged on the vertices of a graph whose edges represent the interactions between neighbors. In particular, the D-Wave quantum annealer implements a graph with the *Chimera topology* (or the more recent *Pegasus topology*) [5,12] and a general problem Hamiltonian $H_{\boldsymbol{\Theta}}$, as defined below in (4), while a transverse external field implements the disorder Hamiltonian H_D as defined in (5).

The general optimization problem that the D-Wave's machine solves is expressed in terms of the *Ising model*, i.e., its solution $\mathbf{z}_0 \in \{-1, 1\}^n$ is given by the minimization of the following cost function:

$$\mathsf{E}(\boldsymbol{\Theta}, \mathbf{z}) = \sum_{i\in V} \theta_i z_i + \sum_{(i,j)\in E} \theta_{ij} z_i z_j \quad \mathbf{z} = (z_1, \ldots, z_n) \in \{-1, 1\}^n \tag{2}$$

where (V, E), with $|V| = n$, is a subgraph of the architecture employing Chimera (Pegasus) topology, the binary variables $z_i \in \{-1, 1\}$ are physically realized by the outcomes of measurements on the qubits located in the vertices, and $\boldsymbol{\Theta}$ is the matrix of the coefficients of E, called *weights*, defined as:

$$\boldsymbol{\Theta}_{ij} := \begin{cases} \theta_i & i = j \\ \theta_{ij} & (i, j) \in E \\ 0 & (i, j) \notin E. \end{cases} \tag{3}$$

The cost function (2) is realized by means of the Hamiltonian of a spin glass:

$$H_{\boldsymbol{\Theta}} = \sum_{i\in V} \theta_i \sigma_z^{(i)} + \sum_{(i,j)\in E} \theta_{ij} \sigma_z^{(i)} \sigma_z^{(j)}. \tag{4}$$

$H_{\boldsymbol{\Theta}}$ is an operator on the Hilbert space $\mathsf{H} = (\mathbb{C}^2)^{\otimes n}$ with $\sigma_z^{(i)} := I_2 \otimes \cdots \otimes \sigma_z \otimes \cdots \otimes I_2$ where I_2 is the 2×2 identity matrix and the Pauli matrix

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the ith tensor factor.

A transverse field gives the kinetic Hamiltonian to realize the quantum fluctuations:

$$H_D = \sum_{i \in V} \sigma_x^{(i)}, \tag{5}$$

where we adopt the same notation of (4) and σ_x is the Pauli matrix:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Given a problem with a binary encoding, the annealer is initialized by a suitable choice of the weights $\boldsymbol{\Theta}$ and a definition of a mapping of the binary variables in the corresponding qubits. The satisfying assignment of the considered problem corresponds to the ground state $|\mathbf{z}_0\rangle$ of the system whose energy eigenvalue is:

$$\mathbf{z}_0 = \underset{\mathbf{z} \in \{-1,1\}^{|V|}}{\operatorname{argmin}} \ \mathsf{E}(\boldsymbol{\Theta}, \mathbf{z}). \tag{6}$$

If the problem presents distinct solutions, then H_P has a degenerate ground state such that the quantum annealing process and a subsequent measurement produce one of the possible solutions. In the case of the real machine, we have also to take into account the repetitions of the annealing and measurement processes. In fact, the annealer returns a sample of solutions which may or not may contain an optimal one, so we also consider the behavior of quantum annealer as an estimation of $\mathbf{z}_0$:

$$\hat{\mathbf{z}}_0 = \widehat{\underset{\mathbf{z} \in \{-1,1\}^{|V|}}{\operatorname{argmin}} \ \mathsf{E}(\boldsymbol{\Theta}, \mathbf{z})} \tag{7}$$

where the right-hand term represents a process that includes the runs of the annealer and the computation of an estimate from their results. For our purpose, this estimation could be either the selection of one of the values or a statistic on the annealer outputs, for example, the mode. Details about QA and its realization by means of spin glasses and its application to solve optimization problems can be found in the literature [6,13,15].

In this work, we propose a learning procedure based on iterated modifications of the Hamiltonian (4) inspired by a nonstandard variation of tabu search to solve optimization problems that cannot be directly mapped in the architecture of a quantum annealer. In particular, we will focus on the *quadratic unconstrained binary optimization* (QUBO) problem that is defined by:

$$\text{Minimize the function} \quad f(\mathbf{z}) = \mathbf{z}^T Q \mathbf{z} \quad \mathbf{z} \in \{-1, 1\}^n, \tag{8}$$

where Q is a $n \times n$ symmetric matrix of real coefficients. The qubit network of a quantum annealer can represent a QUBO problem at the price of restricting the QUBO structure into a graph with less vertices and edges (imposed by the hardware architecture), such an embedding is the subject of intense research activity (e.g., [4]). Our approach is different: Implementing an iterative representation of the objective

function and a variation of tabu search, we can solve a QUBO problem by means of an original utilization of a quantum annealer without the direct representation of the problem into the annealer graph.

The proposed search scheme, named *learning search*, originates from a nonstandard variation of *tabu search*. Let us recall that tabu search is a kind of local search where worse candidate solutions can be accepted with nonzero probability and already visited solutions are penalized [9,10]. In the next section, we describe our general procedure to implement a tabu-inspired search iterating the initialization of a quantum annealer. The considered scheme produces the candidate solutions for the evaluation of the objective function alternating annealing and re-initialization of the weights in order to penalize a collection of already-tested solutions. Generation and testing of candidate solutions realize a sequence of *moves* in the solution space; however, our scheme is not based on a list of forbidden moves like in standard tabu search but rather on a prescription that discourages certain moves and encourages others toward a better representation of the objective function into the annealer architecture giving rise to a learning process. In this sense, we call the matrix which encodes the learning mechanism *tabu matrix*. Let us clarify that the learned object is an initialization of the quantum annealer, given by the weights matrix $\boldsymbol{\Theta}$ and a mapping of the variables into the qubits, such that $\operatorname{argmin}_{\mathbf{z}} \mathsf{E}(\boldsymbol{\Theta}, \mathbf{z}) = \operatorname{argmin}_{\mathbf{z}} f(\mathbf{z})$, where f is the problem objective function. In the present context, the meaning of the term *learning* refers to the guided evolution toward a faithful representation of global optima in the annealer architecture.

In Sect. 3, we propose the algorithms: Algorithm 1 is a general scheme of learning search on a quantum annealer, and Algorithm 2 is a specific version to solve QUBO problems for which we give a convergence proof (Sect. 4) modeling the search as a Markov chain.

2 Learning search on a quantum annealer

In order to introduce the main idea of a learning strategy for local search in the context of quantum annealing, let us consider a simple example of two qubits where $(V, E) = (\{v_1, v_2\}, \{e\})$ and the cost function (energy) is given by:

$$\mathsf{E}(\boldsymbol{\Theta}, \mathbf{z}) = \theta_1 z_1 + \theta_2 z_2 + \theta_{12} z_1 z_2. \tag{9}$$

Moreover, let us also consider an objective function $f : \{-1, 1\}^2 \to \mathbb{R}$ to minimize (not necessary a QUBO problem for now). The heuristic search starts with the random initialization of the weights $\boldsymbol{\Theta}_1$ and $\boldsymbol{\Theta}_2$, and then two annealing processes produce the first two candidate solutions: $\mathbf{z}_g$ and $\mathbf{z}_b$ with $f(\mathbf{z}_g) \leq f(\mathbf{z}_b)$, i.e., $\mathbf{z}_g$ is the current solution. The basic idea is to generate a set of new weights $\boldsymbol{\Theta}'$ for a subsequent run of the quantum annealer such that the *bad candidate* $\mathbf{z}_b = (z_{b1}, z_{b2}, z_{b3})$, i.e., the already discarded solution, is energetically penalized:

$$\begin{cases} \theta'_1 = z_{b1} \\ \theta'_2 = z_{b2} \\ \theta'_{12} = z_{b1} z_{b2}. \end{cases} \tag{10}$$

Once the weights (10) are initialized a new candidate solution is produced. Let us assume that the corresponding value of f is different from $f(\mathbf{z}_g)$, and then we have two bad candidates $\mathbf{z}_b^{(1)}$ and $\mathbf{z}_b^{(2)}$. In this case, the weight initialization for the next run of the annealer will be:

$$\begin{cases} \theta_1'' = z_{b1}^{(1)} + z_{b1}^{(2)} \\ \theta_2'' = z_{b2}^{(1)} + z_{b2}^{(2)} \\ \theta_{12}'' = z_{b1}^{(1)} z_{b2}^{(1)} + z_{b1}^{(2)} z_{b2}^{(2)}. \end{cases} \tag{11}$$

Example 1 (Toy model for two qubits) After the two initial annealing processes, suppose the outputs be $\mathbf{z}_g = (1, 1)$ and $\mathbf{z}_b = (1, -1)$ then $\mathsf{E}(\boldsymbol{\Theta}', \mathbf{z}) = z_1 - z_2 - z_1 z_2$ with:

$$\underset{\mathbf{z}\in\{-1,1\}^2}{\operatorname{argmin}} \mathsf{E}(\boldsymbol{\Theta}', \mathbf{z}) = \{(1, 1), (-1, 1), (-1, -1)\}$$

then assume to perform an annealing process producing the result $(-1, 1)$ satisfying $f(-1, 1) < f(1, 1)$, so we have two bad candidates: $\mathbf{z}_b^{(1)} = (1, -1)$ and $\mathbf{z}_b^{(2)} = (1, 1)$ and a new current solution $\mathbf{z}_g = (-1, 1)$. The weight initialization for the next annealing process will be:

$$\begin{cases} \theta_1'' = 2 \\ \theta_2'' = 0 \\ \theta_{12}'' = 0 \end{cases}$$

so $\mathsf{E}(\boldsymbol{\Theta}'', \mathbf{z}) = z_1$ for which $\operatorname{argmin}\mathsf{E}(\boldsymbol{\Theta}'', \mathbf{z}) = \{(-1, 1), (-1, -1)\}$. If $\mathbf{z}^* = (-1, -1)$ is the global optimum for the objective function f, then the last step of this toy learning search is the initialization of the cost function:

$$\mathsf{E}(\boldsymbol{\Theta}''', \mathbf{z}) = z_1 + z_2 - z_1 z_2$$

attaining the global minimum in $\mathbf{z}^*$.

Since the function E does not encode any information about f, the search sketched in the example above is obviously equivalent to an exhaustive search at this stage. Let us give a first generalization of the argument above considering n qubits. Suppose to have collected a list $\{\mathbf{z}^{(\alpha)}\}_{\alpha\in I} \subset \{-1, 1\}^n$ of discarded candidates, i.e., there is a current solution $\mathbf{z}_g$ such that $f(\mathbf{z}^{(\alpha)}) \geq f(\mathbf{z}_g)$ for any $\alpha \in I$. These elements can be penalized by the following choice of the weights as a direct generalization of (10) and (11):

$$\begin{cases} \theta_i = B \sum_{\alpha\in I} z_i^{(\alpha)} \\ \theta_{ij} = C \sum_{\alpha\in I} z_i^{(\alpha)} z_j^{(\alpha)} \ (i, j) \in E \end{cases} \tag{12}$$

where B and C are positive constants.

Let us define the *tabu matrix* as the $n \times n$ symmetric matrix with integer elements given by (12):

$$S := \sum_{\alpha \in I} [\mathbf{z}^{(\alpha)} \otimes \mathbf{z}^{(\alpha)} - I_n + \mathrm{diag}(\mathbf{z}^{(\alpha)})]. \tag{13}$$

The proposed procedure updates the matrix S when the number of rejected candidates increases (details in Sect. 3), so we consider its equivalent definition by recursion:

$$S := S + \mathbf{z} \otimes \mathbf{z} - I_n + \mathrm{diag}(\mathbf{z}), \tag{14}$$

with S initialized as the null matrix and $\mathbf{z}$ ranging in the list of bad candidates.

In order to encode information of the objective function f into the weights of the energy function E, we introduce a map $\mu : \mathcal{O} \to \mathcal{E}$ from the set of objective functions $\mathcal{O} := \{f : \{-1, 1\}^n \to \mathbb{R}\}$ to the set of energy functions:

$$\mathcal{E} := \{\mathsf{E}(\boldsymbol{\Theta}, \cdot) \mid \boldsymbol{\Theta} \in \mathcal{M}_E\}, \tag{15}$$

where $\mathcal{M}_E$ denotes the set of matrices of the form (3). The general form of μ that we consider is the following:

$$\mu[f](\mathbf{z}) := \mathsf{E}(\boldsymbol{\Theta}[f], \pi(\mathbf{z})), \tag{16}$$

where $\boldsymbol{\Theta}[f]$ is the matrix of weights associated with f according to a certain law and π is a permutation of the variables changing the binary encoding into the qubits of the annealer.

We are interested in the implementation of the learning search by means of the tabu matrix S (13) which encodes the energetic penalization of the rejected candidate solutions. To this end, we need to clarify how the matrix S and the permutation π interact. Let S be the tabu matrix generated by the bad candidates $\{\mathbf{z}^{(\alpha)}\}_{\alpha \in I}$ according to (13) and S_π be the tabu matrix generated by $\{\pi(\mathbf{z}^{(\alpha)})\}_{\alpha \in I}$, if P_π is the matrix of the permutation π then $S_\pi = P_\pi^T S P_\pi$. So the tabu-implementing encoding of f into the cost function can be defined by:

$$\mu[f](\mathbf{z}) := \mathsf{E}(\boldsymbol{\Theta}[f] + \lambda\, S_\pi \circ \mathcal{A}, \pi(\mathbf{z})), \tag{17}$$

where $\mathcal{A}$ is the adjacency matrix of the annealer graph, the Hadamard product $\circ$ maps S_π into a matrix of $\mathcal{M}_E$, and the factor $\lambda > 0$ balances the contribution of the tabu matrix and the weights $\boldsymbol{\Theta}[f]$ that carry information about f. The basic idea of definition (17) is that the additive contribution of the tabu matrix energetically penalizes the already-rejected candidate solutions. The goal of λ is to avoid that the action of the tabu matrix S obscures the information about f; in general, λ is a decreasing function of the number of bad candidates penalized by S.

Once the quantum annealing process has produced an estimate of the global minimum of (17), one must apply π^{-1} on the result to read the solution in the original

variable encoding. During the iterative search, a sequence of μ-encodings is generated to explore the structure of the objective function. We use a permutation of the variables in (16) and (17) for sake of simplicity, but any invertible transformation on $\{-1, 1\}^n$ is eligible. This class includes the permutations composed with local changes of sign; however, it is not clear which advantage these further transformations could give within the considered scheme.

In the specific case of QUBO problems, the objective function is a quadratic form $f : \{-1, 1\}^n \to \mathbb{R}$ with $f(\mathbf{z}) = \mathbf{z}^T Q\mathbf{z}$ where Q is a real symmetric $n \times n$ matrix. The learning strategy and the evolving representation of the problem in the annealer can be summarized as follows: The matrix Q representing the objective quadratic form is *piecewise* mapped into the annealer architecture and *deformed* by means of the tabu matrix to energetically penalize the spin configurations corresponding to solutions that are far from the optimum. Note that in general the current representation of the problem is *not* a subproblem. More precisely, let us define the encoding μ in the QUBO case:

$$\mu[f_Q] := \mathsf{E}(P^T Q P \circ \mathcal{A}, P\mathbf{z}), \tag{18}$$

where P is a permutation matrix of order n and P^T is its transpose. Thus, the action of μ is realized by mapping some elements of Q into the weights, and the mapped elements are selected by P. The tabu-implementing encoding of Q into the annealer induced by the permutation matrix P turns out to be:

$$\begin{aligned} \mu[f_Q](\mathbf{z}) &= \mathsf{E}(P^T Q P \circ \mathcal{A} + \lambda\, P^T S P \circ \mathcal{A}, P\mathbf{z}) \\ &= \mathsf{E}(P^T (Q + \lambda S) P \circ \mathcal{A}, P\mathbf{z}). \end{aligned} \tag{19}$$

In the next section, the form (17) of the tabu-implementing encoding is included in a general scheme for quantum annealing learning search, whose QUBO version uses (19) instead.

3 Hybrid quantum-classical learning search

The idea of penalizing some solutions in the annealer initialization can be exploited for defining quantum-classical hybrid schemes for search algorithms with a learning mechanism of the problem encoding into the quantum architecture. Algorithm 1 shows their general form. The goal is to find the minimum of f with a quantum annealer whose adjacency matrix is $\mathcal{A}$ and its energy function is $\mathsf{E}(\boldsymbol{\Theta}, \mathbf{z})$. The key points of the scheme are definition and updating of the tabu matrix S and the learning of the mapping of the problem variables into the qubits of the annealer architecture.

Initially, two candidate solutions are generated (Algorithm 1, lines 1–6), and while the best is used for a usual initialization (Algorithm 1, line 4), the worst is used for initializing the tabu matrix (Algorithm 1, lines 5–6). In the iterative part of the scheme (Algorithm 1, lines 7–18), new candidate solutions are repeatedly generated and tested, and the candidates that are discarded are used to upgrade the tabu matrix (Algorithm 1, line 16). This permits to obtain the learning mechanism generating

successive encodings $\mu : \mathcal{O} \to \mathcal{E}$ of the objective function into the annealer energy function, and consequently solutions (Algorithm 1, lines 8–9) that take into account: (1) the encoding corresponding to the best solution so far and (2) the tabu matrix scaled by a balancing factor λ. In general, λ is dynamically updated (Algorithm 1, line 17), so it can take into account the number of rejected solutions and/or the actual values of the matrix S. The balancing factor balances the contributions of the tabu matrix and the objective function (Algorithm 1, line 9).

Algorithm 2 instantiates the general scheme of quantum annealing learning search in the QUBO case, and it takes the structure of a simulated annealing with quantum annealer calls. Differently from Algorithm 1, the *encodings* are here completely specified by randomly generated permutations. The *candidate solution generation* combines quantum annealing with a random perturbation of the candidate. The *candidate solution acceptance* allows for random acceptance of suboptimal candidates, and the *termination conditions* are explicitly given. The details of Algorithm 2 are presented below.

Data: Annealer adjacency matrix $\mathcal{A}$ of order n
Input: $f(\mathbf{z})$ to be minimized with respect to $\mathbf{z} \in \{-1, 1\}^n$, annealer energy function $\mathsf{E}(\boldsymbol{\Theta}, \mathbf{z})$
Result: $\mathbf{z}^*$ vector minimum of f
1 randomly generate two maps μ_1 and μ_2 of f: $\mu_1[f](\mathbf{z}) := \mathsf{E}(\boldsymbol{\Theta}_1[f], \pi_1(\mathbf{z})) \equiv \mathsf{E}(\boldsymbol{\Theta}_1, \pi_1(\mathbf{z}))$ and $\mu_2[f](\mathbf{z}) := \mathsf{E}(\boldsymbol{\Theta}_2[f], \pi_2(\mathbf{z})) \equiv \mathsf{E}(\boldsymbol{\Theta}_2, \pi_2(\mathbf{z}))$ where π_1 and π_2 are permutations of the variables;
2 find $\mathbf{z}_1$ and $\mathbf{z}_2$ whose images $\pi_1(\mathbf{z}_1)$, $\pi_2(\mathbf{z}_2)$ estimate the minimum of $\mathsf{E}(\boldsymbol{\Theta}_1, \cdot)$ and $\mathsf{E}(\boldsymbol{\Theta}_2, \cdot)$ in the annealer;
3 evaluate $f(\mathbf{z}_1)$ and $f(\mathbf{z}_2)$;
4 use the best to initialize $\mathbf{z}^*$ and the map μ^*;
5 use the worst to initialize $\mathbf{z}'$;
6 initialize the tabu matrix: $S \leftarrow \mathbf{z}' \otimes \mathbf{z}' - I_n + \mathrm{diag}(\mathbf{z}')$; initialize the balancing factor λ;
7 **repeat**
8 from μ^* and S generate $\boldsymbol{\Theta}[f]$ and π:
9 apply the encoding $\mu[f](\mathbf{z}) := \mathsf{E}(\boldsymbol{\Theta}[f] + \lambda S_\pi \circ \mathcal{A}, \pi(\mathbf{z})) \equiv \mathsf{E}(\boldsymbol{\Theta}, \pi(\mathbf{z}))$ with $S_\pi = P_\pi^T S P_\pi$ where P_π is the matrix of the permutation π;
10 find $\mathbf{z}'$ whose image $\pi(\mathbf{z}')$ estimate the minimum of $\mathsf{E}(\boldsymbol{\Theta}, \cdot)$ in the annealer;
11 **if** $\mathbf{z}' \neq \mathbf{z}^*$ **then**
12 evaluate $f(\mathbf{z}')$;
13 **if** $\mathbf{z}'$ *is better of* $\mathbf{z}^*$ **then**
14 $swap(\mathbf{z}', \mathbf{z}^*)$; $\mu^* \leftarrow \mu$;
15 **end**
16 use $\mathbf{z}'$ to update the tabu matrix S: $S \leftarrow S + \mathbf{z}' \otimes \mathbf{z}' - I_n + diag(\mathbf{z}')$;
17 update the balancing factor λ;
18 **end**
19 **until** *convergence or maximum number of iterations conditions*;
20 **return** $\mathbf{z}^*$;

Algorithm 1: Quantum annealing learning search general scheme

Encodings In Algorithm 2, the tabu-implementing encodings are generated with permutations according to formulas (18) and (19). To this end, we call an additional function $g(P, pr)$ (Algorithm 3) that modifies a permutation by considering for shuffling each element with probability p. The function $g(P, pr)$ produces the permutations that induce the encodings of the objective function into the annealer architecture

Data: Matrix Q of order n encoding a QUBO problem, annealer adjacency matrix $\mathcal{A}$ of order n
Input: Energy function of the annealer $\mathsf{E}(\boldsymbol{\Theta}, \mathbf{z})$, permutation modification function $g(P, p)$, minimum probability $0 < p_\delta < 0.5$ of permutation modification, probability decreasing rate $\eta > 0$, candidate perturbation probability $q > 0$, number N of iterations at constant p, initial balancing factor $\lambda_0 > 0$, number of annealer runs $k \geq 1$, termination parameters i_{max}, N_{max}, d_{min}
Result: $\mathbf{z}^*$ vector with n elements in $\{-1, 1\}$ solution of the QUBO problem

function $f_Q(x)$**:**
  return $x^T Q x$;
$P \leftarrow I_n$;
$p \leftarrow 1$;
$P_1 \leftarrow g(P, 1)$; $P_2 \leftarrow g(P, 1)$; // call Algorithm 3 for generating two permutation matrices
$\boldsymbol{\Theta}_1 \leftarrow P_1^T Q P_1 \circ \mathcal{A}$; $\boldsymbol{\Theta}_2 \leftarrow P_2^T Q P_2 \circ \mathcal{A}$; // weights initialization according to (20)
run the annealer k times with weights $\boldsymbol{\Theta}_1$ and $\boldsymbol{\Theta}_2$
$\mathbf{z}_1 \leftarrow P_1^T \widehat{\operatorname{argmin}}_{\mathbf{z}}(\mathsf{E}(\boldsymbol{\Theta}_1, \mathbf{z}))$; $\mathbf{z}_2 \leftarrow P_2^T \widehat{\operatorname{argmin}}_{\mathbf{z}}(\mathsf{E}(\boldsymbol{\Theta}_2, \mathbf{z}))$; // estimate energy argmin, P_1^T and P_2^T map back the variables
$f_1 \leftarrow f_Q(\mathbf{z}_1)$; $f_2 \leftarrow f_Q(\mathbf{z}_2)$; // evaluate f_Q
// use the best to initialize $\mathbf{z}^*$ and P^*; use the worst to initialize $\mathbf{z}'$
if $f_1 < f_2$ **then**
  $\mathbf{z}^* \leftarrow \mathbf{z}_1$; $f^* \leftarrow f_1$; $P^* \leftarrow P_1$ $\mathbf{z}' \leftarrow \mathbf{z}_2$;
else
  $\mathbf{z}^* \leftarrow \mathbf{z}_2$; $f^* \leftarrow f_2$; $P^* \leftarrow P_2$; $\mathbf{z}' \leftarrow \mathbf{z}_1$;
end
if $f_1 \neq f_2$ **then** $S \leftarrow \mathbf{z}' \otimes \mathbf{z}' - I_n + diag(\mathbf{z}')$; // use $\mathbf{z}'$ to initialize the tabu matrix S
else $S \leftarrow 0$; // otherwise set all the elements of S to zero
$e \leftarrow 0$; $d \leftarrow 0$; $i \leftarrow 0$; $\lambda \leftarrow \lambda_0$;
repeat
  $Q' \leftarrow Q + \lambda S$; // scale and add the tabu matrix
  if N *divides* i **then**
    $p \leftarrow p - (p - p_\delta)\eta$;
  end
  $P \leftarrow g(P^*, p)$; // call Algorithm 3 that returns a modified permutation
  $\boldsymbol{\Theta}' \leftarrow P^T Q' P \circ \mathcal{A}$; // weights initialization according to (21)
  run the annealer k times with weights $\boldsymbol{\Theta}'$
  $\mathbf{z}' \leftarrow P^T \widehat{\operatorname{argmin}}_{\mathbf{z}}(\mathsf{E}(\boldsymbol{\Theta}', \mathbf{z}))$; // estimate energy argmin, P^T maps back the variables
  with probability q $\mathbf{z}' \leftarrow h(\mathbf{z}', p)$; // possibly perturb the candidate by calling Algorithm 4
  if $\mathbf{z}' \neq \mathbf{z}^*$ **then**
    $f' \leftarrow f_Q(\mathbf{z}')$; // evaluate f_Q
    if $f' < f^*$ **then**
      $swap(\mathbf{z}', \mathbf{z}^*)$; $f^* \leftarrow f'$; $P^* \leftarrow P$; $e \leftarrow 0$; $d \leftarrow 0$; // $\mathbf{z}'$ is better
      $S \leftarrow S + \mathbf{z}' \otimes \mathbf{z}' - I_n + diag(\mathbf{z}')$; // use $\mathbf{z}'$ to update the tabu matrix S
    else
      $d \leftarrow d + 1$;
      with probability $p^{(f'-f^*)}$ $swap(\mathbf{z}', \mathbf{z}^*)$; $f^* \leftarrow f'$; $P^* \leftarrow P$; $e \leftarrow 0$;
    end
    update the balancing factor λ with $\lambda \leq \lambda_0$;
  else
    $e \leftarrow e + 1$;
  end
  $i \leftarrow i + 1$;
until $i = i_{max}$ *or* $(e + d \geq N_{max}$ *and* $d < d_{min})$;
return $\mathbf{z}^*$;

Algorithm 2: Quantum annealing learning search for QUBO problems.

(Algorithm 2 lines 5 and 23). The mappings to the parameters of the annealer take into account the actual annealer topology represented by the graph matrix $\mathcal{A}$ (Algorithm 2, lines 6 and 24). The permutations are also used to remap the solutions found by the annealer to the initial space of solutions of the problem (Algorithm 2, lines 8 and 26) and to represent the best map P^*. As an effect of Algorithm 3, if $p = 1$, the resulting permutation is purely random; with $0 < p < 1$, the permutation resembles partially the initial one; if $p = 0$, the output permutation would coincide with the input one. However, this last circumstance does not occur because periodically the probability of an element to be shuffled decreases to the value $0 < p_\delta < 0.5$ with rate η (Algorithm 2, lines 20–22).

Input: A permutation matrix P of order n (p_i row vector of P), a probability pr of an element to be considered for shuffling
Result: A permutation Matrix P'
1 associative map m;
2 **foreach** $i \leftarrow 1 \ldots n$ **do**
3 | with probability pr $m[i] \leftarrow i$; `// select the elements to be shuffled`
4 **end**
5 shuffle map m;
6 **foreach** $i \leftarrow 1 \ldots n$ **do**
7 | **if** $i \in m$ **then**
8 | | $p_i{}' \leftarrow p_{m[i]}$
9 | **else**
10 | | $p_i{}' \leftarrow p_i$
11 | **end**
12 **end**
13 return P';

Algorithm 3: Permutation matrix modification function $g(P, pr)$

Candidate solution generation The matrix Q of the QUBO problem interacts additively with the tabu matrix S scaled by a balancing factor λ (Algorithm 2, line 19) in order to guide the search of the solution by quantum annealing with an energy profile consistent with (19). A crucial effect of this summation is that, in its iterative part, the algorithm does not search anymore for solutions of subproblems as done instead in its initialization. In fact, S contains information about the bad candidates (Algorithm 2, lines 15 and 32) whose objective function values are greater than f^*. The balancing factor λ, initially set to λ_0, is decreased (Algorithm 2, line 37); for example, a sensible choice could be $\lambda \leftarrow \min\left\{\lambda_0, \frac{\lambda_0}{2+i-e}\right\}$ where $2 + i - e$ is the current number of rejected solutions. The candidate solution found by the annealer (Algorithm 2, line 26) is then perturbed with probability q (Algorithm 2, line 27) by the function $h(\mathbf{z}', p)$ (Algorithm 4) in order to guarantee the convergence (see Sect. 4).

Input: A vector $\mathbf{z} \in \{-1, 1\}^n$, a probability pr of a component to be switched
Result: A vector in $\{-1, 1\}^n$
1 **foreach** $i \leftarrow 1 \ldots n$ **do**
2 | with probability pr $z_i \leftarrow -z_i$; `// switch the component`
3 **end**
4 return $\mathbf{z}$;

Algorithm 4: Candidate perturbation function $h(\mathbf{z}, pr)$

Candidate solution acceptance At line 35 of Algorithm 2, a suboptimal solution is accepted with probability $p^{(f'-f^*)}$. By direct comparison with the common rule of acceptation in simulated annealing, namely $p^{(f'-f^*)} = e^{-\frac{(f'-f^*)}{T}}$, it is possible to interpret the role of p by observing that $p = e^{-1/T}$. Namely, in terms of simulated annealing, the parameter p is the probability to accept a suboptimal solution at temperature $T = -\ln^{-1} p$ when the objective function worsens of a unity.

Termination conditions The cycle of Algorithm 2 ends with convergence or when the maximum number of iterations is reached. Line 17 of Algorithm 2 defines three counters for controlling the end of the cycle: e counts the number of consecutive times that a current solution is generated (Algorithm 2, line 39); d counts the number of times the current solution and the new solution differ and the current is better (Algorithm 2, line 34); finally, the variable i simply counts the number of iterations. The counters are compared against input parameters in the termination condition.

Let us illustrate the form that the encodings (16) and (17), included in Algorithm 1, have in Algorithm 2 where they are induced by permutations as in (18) and (19). More precisely, in Algorithm 2 the maps μ_1, μ_2, introduced in the general scheme (Algorithm 1, line 1), are:

$$j = 1, 2 \qquad \mu_j[f](\mathbf{z}) = \mathsf{E}(\mathbf{\Theta}_j, \pi(\mathbf{z})) = \mathsf{E}(P_j^T Q P_j \circ \mathcal{A}, P_j \mathbf{z}). \tag{20}$$

Moreover, in Algorithm 2 the map μ (generated at line 8 in the general scheme of Algorithm 1) is:

$$\mu[f](\mathbf{z}) = \mathsf{E}(\mathbf{\Theta}, \pi(\mathbf{z})) = \mathsf{E}(P^T (Q + \lambda S) P \circ \mathcal{A}, P\mathbf{z}). \tag{21}$$

A comparison between (20) and (21) shows that during the initialization the candidates are found as solutions of subproblems, whereas the problems solved in the iterative part are not in general subproblems of the full problem.

Let us remark that in the real QA setting the weights do not range over all $\mathbb{R}$; a physical machine is characterized by bounded ranges so that: $\theta_i \in [-\delta, +\delta]$ and $\theta_{ij} \in [-\gamma, +\gamma]$, for the D-Wave 2000Q $\delta = 2$ and $\gamma = 1$ [4]. However, the D-Wave system has a built-in scaling function that automatically scales the problem to make maximum use of the available ranges [5], so arbitrary real weights can be used as inputs.

4 Convergence

Algorithm 2 has characteristics, including its simulated annealing structure, that permit to reach conclusions about its convergence. In order to prove that it converges to a global optimum of the considered QUBO problem, we basically apply a result [8] for simulated annealing modeled in terms of Markov processes.

Let us summarize the general result which implies the convergence of our algorithm. Consider a simulated annealing process to find the global minimum of an objective function $F : X \to \mathbb{R}$ defined on a finite set $X = \{x_1, \ldots, x_n\}$. Let A be a stochastic

$(n \times n)$-matrix,[1] called *generation matrix*, defined as follows: if x_i is the current solution, then the element a_{ij} is the probability that x_j is found as a candidate for the next solution. Once a candidate solution is selected, an acceptance probability must be defined. Let us assume that if x_j is a candidate to be the successor of x_i, then it is accepted as the new current solution with probability:

$$\mathbb{P}_{ij}(T) = \begin{cases} 1 & F(x_j) < F(x_i) \\ \mathrm{e}^{-\frac{F(x_j)-F(x_i)}{T}} & \text{otherwise} \end{cases} \tag{22}$$

where T is the decreasing temperature parameter of the simulated annealing. Definition (22) entails that a suboptimal solution has nonzero probability to be accepted and it vanishes as $T \to 0$.

For A and $\mathbb{P}_{ij}$ given, the single steps of the process are described by the transition matrix $M(T)$ whose elements

$$m_{ij}(T) := a_{ij} \cdot \mathbb{P}_{ij}(T) \tag{23}$$

are the probability that if the current solution is x_i, then the next solution will be x_j at temperature T. Modeling a simulated annealing process as a Markov chain, we can state the following definition.

Definition 2 Let $M(T)$ be the transition matrix at temperature T of a simulated annealing process with objective function $F : X \to \mathbb{R}$. The probability distribution π_T on X satisfying

$$\pi_T(x_j) = \sum_{i=1}^{n} \pi_T(x_i) m_{ij}(T) \tag{24}$$

is called *stationary distribution* of $M(T)$.

Assuming the transition matrix $M(T)$ to be irreducible in the sense of Markov chains, i.e., x_i is accessible from x_j for all i, j, and it is aperiodic for any $T > 0$ then $M(T)$ has a unique stationary distribution π_T for any $T > 0$ and $\lim_{T\to 0} \pi_T$ exists [11].

Following [8] a simulated annealing scheme can be explicitly realized by means of a number L of decreasing temperature levels $T_1 \geq \cdots \geq T_L \simeq 0$ and a number N of iterations at each temperature level. So the statistic produced by many iterations at the temperature level T_l reproduces the distribution π_{T_l}. Now let us consider a remarkable result about Markov chains [2,3] in the following proposition.

Proposition 3 *Let $\{M(T)\}_{T>0}$ be a family of transition matrices and $\{T_l\}_{m=1,\ldots,L}$ a set of temperature levels such that $T_1 \geq \cdots \geq T_L \simeq 0$.*

[1] Namely, the elements $\{a_{ij}\}$ of A satisfy: $a_{ij} \geq 0\ \forall i, j$ and $\sum_j a_{ij} = 1\ \forall i$.

If the inhomogeneous Markov chain defined by $\{M(T_l)\}_{l=1,\ldots,L}$ *converges, then its limit distribution corresponds to*

$$\pi^* = \lim_{T \to 0} \pi_T, \tag{25}$$

where π_T *is the stationary distribution of* $M(T)$ *for any* $T > 0$.

As a consequence, the annealing process modeled by the chain is successful, i.e., there is the convergence to one of the global optima, when $\{M(T_l)\}_{l=1,\ldots,L}$ converges and π^* is nonzero only on the global minima of the objective function. About the convergence of the inhomogeneous Markov chain, we have that it converges if the temperature descent is sufficiently slow [2,3], i.e., $N \cdot L \to +\infty$ (a good choice is considered to be $L \simeq 10$ and $N \simeq 10^3$ [8]).

In order to check whether π^* of the considered simulated annealing process recognizes the global optima, we must consider the *neighborhood graph*, i.e., the directed graph $G = (X, \mathfrak{E})$ defined by X as the vertex set and $\mathfrak{E} := \{(x_i, x_j) : a_{ij} > 0\}$ where $A = \{a_{ij}\}$ is the generation matrix defined above. The assumption that $M(T)$ is irreducible for every $T > 0$ is equivalent to the fact that G is strongly connected.[2]

Definition 4 The neighborhood graph $G = (X, \mathfrak{E})$ is said to be *weakly reversible* when for every $\tilde{F} \in \mathbb{R}$ and any $x_i, x_j \in X$ we have that x_j can be reached from x_i along a direct path $(x_i, y_1, \ldots, y_m, x_j)$ such that $F(y_k) \leq \tilde{F}\ \forall k = 1, \ldots, m$ if and only if x_i can be reached form x_j along such a path.

Now let us consider a more general model where the generation matrix may be varying with the temperature parameter; indeed, we have a family of generation matrices $\{A(T)\}_{T>0}$. This is a remarkable feature for the tabu search applications [8] where the generation matrix does not depend only on T but also on the information gained in previous iterations. In [7], there is a crucial result to establish the convergence of a generalized simulated annealing algorithm characterized by a family $\{A(T)\}_{T>0}$ of generation matrices.

Theorem 5 *Let* $\{A(T)\}_{T>0}$ *be a family of generation matrices for a simulated annealing process modeled as a Markov chain within the above scheme.*
If:

(i) *There exists a* $\delta > 0$ *such that* $a_{ij}(T) > 0$ *implies* $a_{ij}(T) \geq \delta$ *for* $i \neq j$ *and* $\forall T > 0$;
(ii) *The neighborhood graph* G *of* $A(T)$ *does not depend on* T;
(iii) G *is weakly reversible;*
then $\pi^*(x) > 0$ *if and only if* $x \in X$ *is an optimal solution.*

We can apply Theorem 5 to prove the convergence of Algorithm 2; in fact, our iterative scheme can be considered a simulated annealing process where the role of temperature parameter is played by the probability p by means of the relation $T = -\ln^{-1} p$. Moreover, the search realized by Algorithm 2 presents the structure of temperature

[2] G is strongly connected if for any pair of vertices x_i and x_j there is a direct path connecting them.

levels with many runs at each level; in fact, the parameter p is constant for N iterations. These cycles provide the convergence of the iterative search modeled as an inhomogeneous Markov chain.

Proposition 6 *Algorithm 2 converges to one of the global minima of* $x^T Qx$.

Proof To prove the statement, we show that (1) the proposed algorithm presents the structure described above and (2) it verifies the hypotheses of Theorem 5.

(1) First of all, let us recall that the generation of a new candidate solution, with a corresponding generation matrix, is described from line 19 to line 27. Moreover, let us check that the acceptation probability is of the form (22), i.e., if $\mathbf{z}^*$ is the current solution, then the probability that $\mathbf{z}'$ is accepted as the new current solution has the following form:

$$\mathbb{P}(\mathbf{z}^*, \mathbf{z}', T) := \begin{cases} 1 & f' < f^* \\ \mathrm{e}^{-\frac{f'-f^*}{T}} & \text{otherwise} \end{cases} \tag{26}$$

where $f' = f(\mathbf{z}')$ and $f^* = f(\mathbf{z}^*)$. The acceptation probability of our search corresponds to (26) by posing $T = -\ln^{-1} p$ as shown from line 30 to line 36 of the algorithm. The temperature lowering of this simulated annealing process is described by line 21.

For a fixed value of p, we can identify a single step of the iterative search from line 19 to line 41 that can be described by a transition matrix $M(p)$. The *repeat-until* and the *if* statement of lines 20–22 realize the structure of temperature levels, so the corresponding inhomogeneous Markov chain converges as $i_{max} \to +\infty$.

(2) Let us initially focus on hypothesis (iii) of Theorem 5. It suffices to show that if a solution was reached from another solution, then it is also possible to do the contrary because this implies weak reversibility. By denoting with s the number of components that are equal between the two solutions, the probability to generate again the previous solution is $q \cdot (1-p)^s \cdot p^{n-s}$ as an effect of line 27 and Algorithm 4. Since $p_\delta < 0.5$, we have that $q \cdot (1-p)^s \cdot p^{n-s} \geq q \cdot (\eta p_\delta)^n$. This last observation leads to see that also hypothesis (i) hold, in fact $q \cdot (\eta p_\delta)^n$ corresponds to $\delta > 0$ of Theorem 5, for q, η and p_δ strictly positive.

The above observation holds indeed for any pair of solutions. Consequently, hypothesis (ii) is also satisfied because the neighborhood graph associated with the generation is complete for any p; in fact, from any current solution $\mathbf{z}^*$ there is a probability larger than $q \cdot (\eta p_\delta)^n$ to reach any other solution. □

Let us make apparent how an aspect of the proposed quantum annealing learning search does not have any consequence on its convergence. Describing Algorithms 1 and 2, we have not assumed that the quantum annealer always returns the optimal solution to the annealer energy function (6) in favor of the realistic assumption that the annealer returns a sample of solutions which may or not may contain an optimal one (7). However, even if the estimation of the ground state solution in (7) and used in lines 8 and 26 of Algorithm 2 was perfect, namely it coincided to (6), the algorithm converges anyway. In fact, there is always a non-null probability that a non-optimal

solution is accepted (Algorithm 2, line 35). On the other hand, the algorithm converges also when the quantum machine returns a poor estimation of a global minimum of the annealer energy function most of the time. In this case, the procedure reduces to classical simulated annealing with random generation of the candidate solutions whose convergence is guaranteed [8].

Note that Proposition 6 holds for any QUBO problem, including also those whose solution does not contain solutions of subproblems, namely a violation of the so-called *optimal-substructure property*. In fact, our algorithm does not decompose the problem in subproblems but solves a series of problems defined by the additive interaction between the weights that carry information about f and the tabu matrix. In general, this series is not composed by subproblems of the full problem consistently with the possible absence of the optimal-substructure property.

A crucial aspect for convergence of Algorithm 2 is the probabilistic behavior of the scheme. Let us point out where quantum and classical probabilities intervene in the iterative part of the algorithm: (1) Quantum probability in the output of the annealer (Algorithm 2, lines 25 and 26); (2) Classical probability p of permutation modification (line 23); (3) Classical probability q of solution perturbation (line 27); (4) Classical probability $p^{f'-f^*}$ of acceptance of suboptimal solutions (line 35). The probabilities (1–3) characterize the process of solution generation whereas (4) influences the solution acceptance probability. As shown in the proof of Proposition 6, classical probabilities p and q allow for a nonzero probability to reach any candidate solution; this fact is the keystone of the convergence argument.

The proof of Proposition 6 makes evident that our algorithm presents the structure of a simulated annealing that delegates part of the search to a quantum annealer. However, the quantum annealer does not search purely on a representation of the objective function. In fact, we stress that the tabu-implementing encoding produces its effects in the quantum part of the scheme. Thus, the presented algorithm cannot be reduced to a simple wrapping of quantum annealer calls in a simulated annealing procedure. This implies that the power of the quantum annealing acts on a mixed representation of the objective function and of the current state of the search that includes the bad elements arranged in the tabu matrix. This circumstance may give computational advantages that must be further analytically and empirically investigated.

Let us clarify that the given proof ensures the convergence asymptotically as the number of iterations goes to infinity. Empirical evaluation is needed to elucidate the actual number of iterations required in practice for each class of problems. It is interesting to note that Proposition 6 also shows a constructive way of finding values of parameters $\boldsymbol{\Theta}$ that encode the very same minima of the QUBO problem. This means that it would be possible for a class of QUBO problems to run several times the algorithm on different instances and learn the encoding into the annealer architecture from examples. In this way, the learned encoding could be used to initialize other instances of the same class of problems decreasing the required number of iterations.

5 Related work

Tran et al. [17] propose a hybrid quantum-classical approach that includes a quantum annealer. In their proposal, the classical part of the algorithm maintains a search tree and decides which of two solvers, a quantum annealer and classical one, to call on a subproblem. Notably, they implemented their approach on a D-wave quantum annealer and their goal is to realize a complete search. Earlier proposals of quantum-classical schemes include the work of Rosenberg et al. [16] who proposed to decompose a big problem in subproblems that can be solved by the annealer and validated their approach using simulated annealing. More recently, Abbott et al. [1] suggested that the search for the embedding of the problem into the annealer topology can be critical in the quest to achieve a quantum speedup for an hybrid quantum-classical algorithm. The D-wave handbook [5] reports at page 74 that their *QSage* package implements a hybrid tabu search algorithm for problem decomposition for the annealer. Moreover, D-wave also released the software *qbsolv* that allows for a decomposition of large QUBO problems.

Our approach differs from what previously published in two fundamental ways: (1) we inserted the tabu mechanism inside the annealer stage by means of the matrix S and (2) we provide a proof of convergence by mapping the classical part to results about SA procedures.

6 Conclusion and future work

We have presented a general scheme for the application of a nonstandard version of tabu search to define a learning search, a specific instance of a hybrid quantum-classical algorithm, and proved its convergence.

The general strategy of the proposed algorithm is to implement a search equipped with a learning mechanism of the problem encoding into the quantum annealer architecture. The procedure is essentially based on: (1) defining and updating a tabu matrix that deforms the weights of the annealer energy function; (2) an evolving mapping of the variables into the qubits. In Algorithm 2, the learning of the encoding is determined by the generation of permutations depending on a probability that can be interpreted as the decreasing temperature of a simulated annealing process. Applying some results about convergence of SA processes modeled by Markov chains, we proved the convergence of Algorithm 2 to a solution of the QUBO problem.

The main direction for further investigation is the estimation of the run time of the presented algorithm (Algorithm 2) and its empirical validation. Let us point out that if we choose to construct 10 temperature levels with 10^3 iterations for each level, we need to call the annealer $k \cdot 10^4$ times, where k is the number of annealer calls per iteration, with an annealing time for a single anneal of 20μs (D-wave 2X [17]). However, our algorithm improves the representation of the objective function into the annealer during the search, and thus, we can conjecture a gain in the number of annealer calls and a consequent speedup of the optimization.

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